CODIMENSION OF THE LOCAL RING AT A DECENT POINT EQUALS THE CODIMENSION IN THE RESIDUAL SPACE

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Editorial note. The uncarded ordered-bounds input is declared external.

Allowed standard prerequisites (editorial).

- Weak normalization and ordered bounds. lemma-bound-weakly-normal (same chapter, NOT carded, declared external). The ordered bounds used in the identification.
- Residual space structure. definition-residual-space (same chapter, structural). The closed subspace t({x}).

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02IZ: focused-seeds/geometry/S9-decent-codimension-local-ring/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 03K6: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2915–2915. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NA: focused-seeds/geometry/S9-decent-codimension-local-ring/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02I3: focused-seeds/geometry/S9-decent-codimension-local-ring/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $T \subset |X|$ be an irreducible closed subset. Let $\xi \in T$ be the generic point (Proposition 03K6). Then $\mathrm{codim}(T, |X|)$ (Topology, Definition 02I3) is the dimension of the local ring of X at ξ (Properties of Spaces, Definition 04NA).*

Proof. Choose a scheme U, a point $u \in U$, and an étale morphism $U \to X$ sending u to ξ. Then any sequence of nontrivial specializations $\xi_e \rightsquigarrow \ldots \rightsquigarrow \xi_0 = \xi$ can be lifted to a sequence $u_e \rightsquigarrow \ldots \rightsquigarrow u_0 = u$ in U by Lemma 03IL. Conversely, any sequence of nontrivial specializations $u_e \rightsquigarrow \ldots \rightsquigarrow u_0 = u$ in U maps to a sequence of nontrivial specializations $\xi_e \rightsquigarrow \ldots \rightsquigarrow \xi_0 = \xi$ by Lemma 03K5. Because $|X|$ and U are sober topological spaces we conclude that the codimension of T in $|X|$ and of $\overline{\{u\}}$ in U are the same. In this way the lemma reduces to the schemes case which is Properties, Lemma 02IZ. □